University of Essex

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Project Proposal Submitted as part of

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Developing a Machine Learning Pipeline for Understanding Well-being in Climate Change Literature

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Abstract

Climate change is far more than an environmental crisis. It has a profound impact on human well-being, altering everything from health — physical and mental — to livelihoods and social stability. Understanding these impacts is essential for creating effective solutions and policies. Climate change research has expanded quickly, but the evidence linking climate impacts and responses to human well-being is patchy across disciplines and often hard to map at scale. In this project, we are proposing a machine learning (ML) pipeline that automatically (i) detects whether a climate-related publication has substantive well-being content and (ii) assigns a well-being theme out of 12 different climate well-being categories to these publications where the theme annotation is meaningful. This proposal expands on prior work done with transformer-based language models (e.g., BERT-style encoders) and addresses extreme class imbalance that is prevalent in evidence-mapping tasks. The planned scheme uses a hierarchical (two-stage) design: a binary gate for Accept vs Reject (well-being), with the first being a binary relevance function (i.e., relevant vs. not), followed by a conditional multi-class theme classifier only used for accepted items. A suite of imbalanced-aware strategies (e.g., class weighing, focal loss, augmentation, cost-sensitive learning, ensemble supervised learning, and contrastive representation learning); a hierarchical approach with the DeBERTa model; and a one-vs-all multimodal approach will be comprehensively studied and compared under rigorous evaluation with a mix of protocols (stratified cross-validation, threshold tuning, and imbalance-aware metrics like macro-F1 and PR-AUC) to select among approaches. The three research approaches have yielded the primary and baseline accuracies of 88%, 91%, and 40%, respectively, which will be monitored and developed later. The expected outcome is a reproducible end-to-end pipeline able to produce auditable predictions, confidence intervals, and rationale to support climate impact and well-being mapping studies suitable for downstream reviews.

Keywords: Climate change literature; well-being; evidence mapping; transformer models; class imbalance; hierarchical classification; DeBERTa; ClimateBERT.

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1 Introduction

1.1 Motivation: The Climate-Wellbeing Evidence Synthesis Challenge

Climate change is now characterized not merely as an environmental issue but overwhelmingly as a defining human condition with essential implications for health, livelihoods, security, and equity (IPCC 2022a). The sixth assessment of the Intergovernmental Panel on Climate Change (IPCC) explicitly acknowledged that climate action, both mitigation and adaptation, interacts with well-being in complex ways: while some policies foster human flourishing via their co-benefits, e.g., improved air quality and green jobs, others produce trade-offs or distributional inequities (IPCC 2022b). Mapping climate-wellbeing linkages is critical for developing equitable climate actions consistent with the Paris Agreement and Sustainable Development Goals.

The scientific literature bridging this nexus has exploded. Callaghan et al. (2025), for example, reported 84,990 peer-reviewed publications which cited climate policy instruments alone and showed that only 0.33% were classified as evidence synthesis studies—a glaring "synthesis gap." Likewise, Callaghan et al. (2021) mapped more than 102,000 climate impact studies. This poses a fundamental challenge: there are no longer enough resources to manually screen tens of thousands of abstracts and journal articles to identify relevant publications, let alone classify them into various thematic taxonomies (Callaghan et al. 2024; van de Schoot et al. 2021).

The well-being factor adds further complication. Evidence mapping needs to establish *which* such themes are being addressed, ranging from mitigation co-benefits and SDG interactions through to equity and well-being metrics. The frequencies of each theme may vary drastically, from hundreds of papers overexpressing certain themes to rare ones (e.g., indigenous well-being vs. mental health) only having 3–10 labeled examples. This results in the **extreme imbalance challenge**: not only is the positive class a minority (5-15% of screened abstracts), but also relevant papers have extreme skewed distribution of themes (ratios up to 28:1); thus, standard classification will suffer from **model collapse**, i.e., predict only the most common themes.

1.2 Progress and Gaps in Climate NLP

Language models based on transformers have radically altered the processing of climate text. Domain-adaptive pretraining on 2M+ climate paragraphs leads to considerable gains over vanilla transformer approaches such as Transformer-BERT with F1 $\approx$ 0.991 for climate classification, F1 $\approx$ 0.838 for sentiment, and F1 $\approx$ 0.757 for fact-checking tasks (Webersinke et al. 2022). Extending this work, Callaghan et al. (2021) screened 102k studies with DistilBERT (F1 0.71, ROC-AUC 0.92), Sietsma et al. (2024b) achieved relevance screening on N = 50,462 abstracts by an aggregated F1 92.2%, and Singh et al. (2024) provided NDC classifiers through macro F1 $\approx$ = 0.76 across sectors.

However, critical gaps persist. They mostly focus on binary or balanced multi-class tasks. Sietsma et al. (2024b) observed performance on imbalanced category classifiers was weaker, showing between 45 – 75% F1 for the rarest labels, and Sietsma et al. (2024a) said that ML models tend to reproduce rather than find new patterns, ultimately manifesting as undercovered topics. Newer work on LLMs is promising but limited: Larosa et al. found 80% agreement with trained expert labels while also noting sensitivity to specific prompt construction and lack of nuance for political speech. The Climate-Eval benchmark (Kurfalı et al. 2025) found that many tasks still prove challenging, often scoring below 0.34 macro-F1 even under few-shot prompting conditions.

Three key gaps remain: (1) Extreme multi-label imbalance (ratios 10:1 to 100:1 and only 3–10 examples per class) causes softmax classifiers to collapse to predicting just 1–3 frequent classes, and (2) Threshold

optimization and calibration are seldom addressed. At the same time it Callaghan et al. (2024) showed that high 5th-percentile recall of 53% is possible with informal heuristics; however, despite neural models being overconfident (Guo et al. 2017), calibration metrics have thus far been underreported. (3) There are optimal behaviors for which different operations become ranking strategies and classification architectures, whereas two-stage versus one-vs.-all performance in extreme imbalance contexts or differences between error propagation and ensemble calibration trade-offs also remain unexplored.

1.3 Research Objectives and Experimental Design

This thesis bridges these gaps with a systematic exploration of classification techniques for climate-wellbeing thematic mapping in the face of extreme class imbalance. We utilize a manually annotated dataset of 1,719 climate policy abstracts that are labeled as either relevant (Accept: 199, 11.58%; Reject: 1,520, 88.42%) or irrelevant and also mapped to one of 12 well-being themes for accepted articles. There is a significant imbalance within the dataset: between 199 accepted papers, the distribution by theme can be as few as 84 samples ("Climate change mitigation options & well-being") up to just 3 samples ("Systems Thinking"), generating an imbalance of 28:1.

We will compare two approaches. The **standard and hierarchical two-step** strategy uses a binary Accept/Reject classifier followed by a 12-class softmax theme classifier. This experienced initial model success, predicting 11 out of 12 themes, using aggressive oversampling and class-weighted loss. This led us to a **one-versus-all (OvA)** approach, creating 12 separate binary classifiers (one per theme) to predict "Is it Theme X? (Accepted)" versus ‘ "All others." That results in 36 total models with 3-fold cross-validation. Across classifiers the following are used: (1) aggressive oversampling (small themes saturated to a minimum of 30, maximum augmentation to 10x); (2) balanced undersampling (3:1 negative–positive ratios); (3) class-balanced loss with weights specific to each theme capped at a value of 10; and finally, (4) per-theme optimization via precision-recall curve analysis on resulting predictions. We train all models using DeBERTa-v3-base (He et al. 2021) and AdamW optimization.

1.4 Key Findings and Contributions

Our analysis uncovers successes as well as major failure modes. The OvA approach **effectively avoided primary model collapse**, leading to full coverage of themes (12/12 themes predicted vs. 11/12 in baseline). However, intial out-of-fold (OOF) evaluation on training data revealed **catastrophic precision loss**: for training OOF metrics precision 14.93%; recall 87.94%; and F1 25.53% was being used to classify Accept class. This led to an initial confusion matrix showing that 65.6% of true rejects had incorrectly been predicted as Accept (997 out of 1,520 rejected papers). On the test set (10,175 papers), the ensemble predicted 47.23% as accepted vs 11.58% of expectation—a 4.1% inflation.

Somewhat surprisingly, analysis of theme distribution indicated an **inverse relationship between training data size and test prediction frequency**. In the case of two relatively low prevalence from 3-7 training samples ("Integrated Responses and Well-being": 6 samples; "Better well-being Metric": 7 samples), they became the most over-predicted, capturing 37.5% and 15.2% of test accepts, respectively versus 3.0% and 3.5% in training.) On the other hand, moderately common themes (20 to 31 samples) were highly under-predicted: "Empowering People & Governance" with its 31 training samples (15.7% of training distribution), fell down to very low 0.0% test predictions (having a single paper), and "Implemented Mitigation Options," which also has 21 samples in the training data present a drop-down data to only count for 10.6% reaching just 0.6%.

By systematically applying root cause analysis we found four compounding factors: (1) **Threshold selection bias**—since both the global threshold ($\tau_{\text{global}} = 0.3$) and per-theme thresholds were optimized on small cross-validation folds maximizing for F1, which favors recall by definition, our qualitative

post-hoc inspection of clustering performance suggested an increase to 0.6-0.7 would have reduced acceptance from 47% to ~10-15%. (2) **Over-sampling artifacts**, 10-repeat copies for rare themes distracted models that memorized lexical patterns instead of robust semantics. (3) **Rare theme overfitting** Classifiers had learned to associate rare themes (e.g., the word "well-being" occurring often in spurious ways among many papers) with positive classes based on only 3–7 examples. (4) **No global rejection constraint**, every theme classifier runs independently, leading to an "any vote wins" case.

Our contributions are threefold. **Methodologically,** we show that one-versus-all classification leads to specific fusion of models, avoiding collapse in extreme levels of multi-class imbalance (28:1); it also implies a lack of F1 maximization on small imbalanced folds while leading to precision-constrained threshold optimization. **Empirically**, we demonstrate via out of fold (OOF) evaluation that excessively aggressive oversampling with 3-10 examples creates an inverse relationship where the rarest themes are shifted to be the most overpredicted, exposing a deep universe and fundamental limits of data augmentation in extreme scarcity. As we observed concerning OOF predictions during training, 85% are false positives, where positive predictions were made when there was no positive label indicating the need for dedicated validation schemes and proper indicator of precision-recall trade-offs rather than accuracy or F1 values. **Practically**, we identify specific rectification approaches: (1) post-hoc threshold recalibration via precision-constrained optimization; (2) temperature scaling (Guo et al. 2017) to mitigate overconfidence; (3) using a dedicated binary Accept/Reject gating classifier to weed out false positives before attaching themes, and (4) investigating few-shot learning techniques (prototypical networks (Snell et al. 2017), meta-learning), or LLM-based sample generation for themes with fewer than 10 examples.

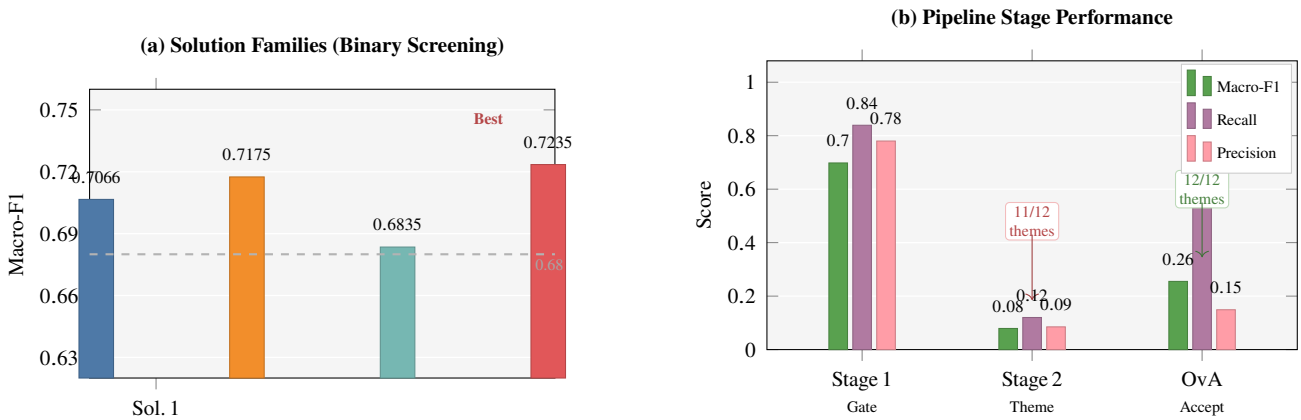

Figure 1: **(a)** All four imbalance-aware strategies exceed macro-F1 > 0.68 on binary screening; Solution 4 (Contrastive+Mixup) is best at 0.724. **(b)** The hierarchical pipeline's Stage 2 theme classifier predicted 11/12 themes (macro-F1 = 0.079), whereas the proposed OvA ensemble recovers full 12/12 coverage, albeit with low precision (0.149), motivating precision-constrained recalibration as future work.

We will seek to not build a production-ready system but bear out experimentation-centered analysis that finds defined failure modes, suggests grounded remediation approaches, and increases what climate NLP knows about the potential and limits of machine learning in extreme imbalance contexts.

2 Related works

This section reviews research relevant to (i) Climate-domain text mining, (ii) Well-being measurement and thematic evidence mapping, (iii) Methodological foundations for imbalanced and hierarchical text classification, and (iv) A brief literature review.

2.1 Climate-domain NLP and literature mining

A growing body of work applies transformer models to climate-related classification and retrieval tasks, including stance detection, policy categorization, and scientific document screening (Devlin et al. 2019; Liu et al. 2019). Domain adaptation is effective in specific domains: ClimateBERT shows that continued pretraining on climate corpora can help, offering a foundation for generalization to downstream tasks (Webersinke et al. 2022). These results encourage testing of both general-purpose transformers and climate-adjusted checkpoints in well-being mapping.

2.2 Well-being concepts and evidence mapping

Well-being is multi-dimensional (e.g., health, social connection, security, and subjective life satisfaction) and is operationalized differently across disciplines and policy frameworks. International frameworks like the Sustainable Development Goals (SDGs) and OECD well-being initiatives provide structured lenses to see connections between climate action and human outcomes (United Nations 2015; OECD 2020). Each evidence mapping in this domain often has (a) relevance screening and (b) thematic assignment, which follows a hierarchical decision process quite naturally.

2.3 Imbalanced and hierarchical text classification

Screening tasks are prone to imbalanced classification, where the majority of items are "non-relevant" ones. Standard cross-entropy can bias a classifier toward the majority class; class weighting and focal loss have been shown to improve minority detection by increasing penalty for rare-class errors, specifically encouraging focus on hard examples (Lin et al. 2017). Data-level techniques refer to methods such as oversampling and the generation of synthetic data, e.g., SMOTE (Chawla et al. 2002)(applied with care for text). Stochastic regularization (e.g., dropout-based uncertainty (Gal and Ghahramani 2016)), stochastic weight averaging (Izmailov et al. 2018), and ensembles that decrease prediction variance (Dietterich 2000) can increase model-level and training-time robustness. For representation learning, mixup-style interpolation (Zhang et al. 2018) and supervised contrastive learning (Khosla et al. 2020) can facilitate class separation and generalization. Hierarchical classification (after a gating decision, then conditional sub-classification) is appropriate for use where only coarse-grained labels are available for much of the data and to combat error propagation through explicit modeling of conditional label structure.

2.4 Literature Review

Author	Main Technique	Result	Key Finding
Callaghan et al. (2025)	Transformer-based ClimateBERT for automated literature classification by policy instrument, sector, and geography	Classified 84,990 publications; identified only 0.33% as evidence synthesis studies	Revealed substantial evidence synthesis gap and mismatches between research attention, policy density, and emissions profiles across sectors

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Author	Main Technique	Result	Key Finding
Callaghan et al. (2021)	Fine-tuned DistilBERT classifier with nested cross-validation for climate impact study screening	F1 score ≈0.71, ROC-AUC ≈0.92 on 102,000+ publications	Potentially attributable anthropogenic impacts occur across 80% of the world's land area; there is a significant attribution gap between high- and low-income regions
Webersinke et al. (2022)	Domain-adaptive pretraining of DistilRoBERTa on 2M climate paragraphs (ClimateBERT)	F1 ≈ 0.991 (climate classification), 0.838 (sentiment), 0.757 (fact-checking); 46-48% reduction in masked-LM loss	ClimateBERT significantly outperforms general-domain models on climate-specific NLP tasks but shows geographic and gender biases
Garrido-Merchán et al. (2023)	Fine-tuning BERT and ClimateBERT on the ClimaText dataset for climate-related financial risk detection	ClimateBERT: 93.5% accuracy vs 92.5% for BERT on 10-K filings	Domain-specialized models (ClimateBERT) achieve statistically significant improvements over general BERT for climate disclosure analysis
Grasso et al. (2024)	BERT-family transformers (BERT, RoBERTa, DistilRoBERTa, ClimateBERT) on manually annotated Twitter dataset (EcoVerse, 3,023 tweets)	Accuracy: 89-90% (eco-relevance), 79% (environmental impact), 81% (stance detection)	Models perform well on eco-relevance but struggle with neutral impact distinction; environmental discourse extends beyond climate change alone
Shyrokykh et al. (2023)	Comparison of lexicon-based, ML (logistic regression, SVM, random forest), and DL (CNN, LSTM) methods on climate tweets	F1: 65% (imbalanced), >96% (balanced); traditional ML matches DL performance with faster training	Simpler ML models (logistic regression, random forest) are sufficient for short-text classification; deep learning offers minimal gains

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Author	Main Technique	Result	Key Finding
Singh et al. (2024)	Fine-tuning sentence-embedding model (BAAI bge-base-en-v1.5) and ClimateBERT on CPo-CD dataset for NDC classification	Macro F1: 0.76 (sectors), 0.84-0.85 (targets/actions), 0.91-0.92 (GHG/mitigation-adaptation)	Generic sentence-embedding models slightly outperform ClimateBERT on policy document classification, enables systematic tracking of Paris Agreement alignment
Katsadaki and Kokla (2024)	Multinomial logistic regression via Amazon Comprehend for multiclass climate report classification (locations, disasters, impacts, time periods)	Accuracy/F1: 0.98 (locations, impacts), 0.90 (disasters), 0.80 (time periods)	Temporal classification from narrative text is hardest (F1 0.80); automated classification reveals patterns in climate risk distribution across space and time
Sietsma et al. (2024b)	ClimateBERT-based relevance classifier + topic modeling (105 topics) for adaptation policy literature mapping	F1 92.2% (relevance); identified 8,691 adaptation articles from 50,462; category classifiers F1 45-75%	Annex I countries dominate (54.3%), the USA alone 30%; this reveals global imbalance in adaptation research and policy instrument patterns (finance internationally, legislation in Europe)
Larosa et al. (2025)	ClimateBERT + Gemini 1.0 LLM with prompt engineering for NDC-SDG mapping and SDG interlinkage detection	80% agreement with expert labels (KnowSDGs); LLMs detect more SDG pairs than expert protocols	LLMs can scale policy analysis, but outputs are sensitive to prompts, show embedded biases, and require expert interpretation; they need human-aware workflows
Kurfalı et al. (2025)	Evaluation of 9 open-source LLMs (2B-70B params: Gemma-2, Qwen-2.5, Llama-2/3, ClimateGPT, Mistral) on 25 climate NLP tasks	Few-shot lifts macro-F1 from 0.45 to 0.53 (+0.09); many tasks remain difficult (<0.34 macro-F1); ClimateGPT gains modest (0.05)	Climate-specialized models show inconsistent advantages; many climate NLP tasks remain challenging even for large models

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Author	Main Technique	Result	Key Finding
Fard et al. (2022)	Domain adaptation of ClimateBERT and diseaseBERT variants to the climate-health corpus (16,078 papers) with vocabulary extension	Proposal stage: expected improvements on NER, binary classification, and fact-checking tasks	Multi-domain pretraining can capture climate-health relationships; it requires integration of climate and medical domain knowledge
Callaghan et al. (2024)	Review and synthesis of ML-assisted screening stopping criteria for systematic reviews	Naïve heuristics achieve 5th-percentile recall as low as 53%; proposed statistical rules show promise but limited validation	Current stopping rules risk missing substantial relevant studies; robust, statistically grounded criteria with explicit confidence estimates are needed.
van de Schoot et al. (2021)	Active learning framework (ASReview) for ML-assisted systematic review screening with multiple classifiers	An open-source tool enables efficient screening; work saved varies by dataset and model	ML can accelerate systematic reviews but requires transparent methodology, validation, and careful stopping criteria

3 Background

3.1 Project context

The project is situated within climate impact and well-being evidence mapping. The operational setting is a two-file workflow, as datasets are common in systematic screening:

- A **human-labeled training file dataset** containing abstracts with a binary screening label (Accept/Reject) and a theme label for accepted items, consisting of 199 accepted papers and 1520 rejected papers. The accepted papers are annotated in 12 different climate well-being categories by professional annotators
- A **large master file** containing nearly 10,000 unlabelled abstracts requiring screening and theme assignment.

Early tests utilized a labeled corpus of 1.7k abstracts with a very strong Accept/Reject skew (minority Accept) and also on a much smaller set of items labeled by theme (accepted items only). This ends up producing the final CSV with predicted Accept/Reject and a theme for accepted predictions, along with a confidence score of the stage-1 decisions. These outputs are designed to assist reviewers by prioritizing potentially relevant papers and summarizing theme distributions for quicker scoping.

3.2 Technical review

Transformer encoders

This research will use transformer encoders (BERT-family) as the backbone model. BERT (Devlin et al. 2019) and RoBERTa (Liu et al. 2019) can be seen as general-purpose baselines that have been improved by self-supervised pretraining methods on the sequence classification task; DeBERTa-style models have further improved such attention representations in recent years, yielding state-of-the-art results on classification tasks (He et al. 2021). We will assess climate-adapted variants (e.g., ClimateBERT) for their possible domain advantages, such as ClimateBERT (Webersinke et al. 2022).

Imbalance-aware learning

The pipeline will adopt both data-level and loss-level techniques:

- **Class weighting** to give a higher penalty to minority class errors.
- **Focal loss** to bring attention to really hard and ambiguous samples (Lin et al. 2017).
- **Augmentation** for text (cautiously manipulated to retain meaning) and SMOTE generalisation of synthetic sampling adapted for sentence representations (Chawla et al. 2002).
- **Ensembles** to stabilize predicted probabilities (Dietterich 2000) across folds/models.

Evaluation under class imbalance

In an imbalanced setting, accuracy may be contaminated by bias. Macro-F1 reflects balanced performance across many classes, and precision-recall analysis is frequently more informative (Saito and Rehmsmeier 2015). Hence, the evaluation plan focuses on macro-F1, class-wise precision/recall/F1, PR-AUC (for the Accept class), and calibrated confidence reporting.

4 Problem Statement

Given its breadth, climate change literature is also a heterogeneous mix of papers, many with no direct connection to human well-being. Manual screening and thematic coding are time-consuming, labor-intensive processes that are subject to ambiguous reviewer interpretations. The main task is to construct a solid ML pipeline, which should be able to:

1. **Screen** climate-related abstracts for well-being relevance (Accept/Reject) with high minority-class recall and reliable confidence.
2. **Map** accepted abstracts to a well-being theme taxonomy (realizing that theme labels are only ascribed for this subset).

The pipeline must handle extreme class imbalance, noisy natural language, and domain shifts of time, journals, and subdisciplines whilst remaining sufficiently interpretable to allow downstream human auditing.

4.1 Research questions

- **RQ1:** How well do transformer-based models predict well-being-related climate abstracts with a high degree of class imbalance?
- **RQ2:** What is the end-to-end usefulness of a hierarchical two-stage design (Accept/Reject gate (→) theme assignment) versus a single multi-task model?
- **RQ3:** What is the performance and reliability trade-off corresponding to different imbalance-handling strategies (weighted loss, focal loss, augmentation, cost-sensitive training, ensemble, contrastive/mixup)?
- **RQ4:** In the context of real evidence-mapping workflows, how do confidence scores and threshold policies need to be calibrated?

5 Goals

5.1 Objectives

This is the groundwork for building and verifying a comprehensive machine learning pipeline that can be used to alert and map climate change literature as it relates to various aspects of well-being. Specific objectives are the following:

1. Build a reproducible preprocessing pipeline (cleaning, normalization, split management, and audit trails) for abstract text and metadata.
2. Implemented and compared multiple model families and training strategies (e.g., baseline transformer, imbalance-aware loss, augmentation, ensembles, contrastive/mixup).
3. Build a hierarchical inference pipeline that outputs CSV with final Accept/Reject class, conditional theme predictions, and confidences.
4. Asking to record a compact evaluation suite for imbalanced screening (macro-F1, PR-AUC, calibration, and error analysis) and theme classification (macro-F1, per-class recall, and taxonomy coverage).
5. Write explicit documentation, ablation results, and a project report that is acceptable for submission as a dissertation.

5.2 Scope and Limitations

Scope: This proposal targets abstraction-level classification of English-language climate literature. The main deliverable is a screening + theme mapping pipeline that can be used for evidence mapping and scoping reviews.

Limitations: (i) full well-being details will not be given in abstracts only; (ii) theme labels are limited and highly skewed for the accepted items; (iii) definitions for labels may vary between annotators, as well as datasets used during model training and evaluation; (iv) there will be limitations on sizes of models trained or number of hyperparameter searches due to computational capacity (e.g., GPU/TPU memory).

Table 2 positions the proposed methodology against a curated selection of prior work across six methodologically critical dimensions. Unlike preceding studies, which individually address at most two or three of these challenges, the proposed pipeline unifies all six capabilities within a single reproducible framework. Notably, no existing work simultaneously handles extreme class imbalance at theme-level granularity, performs hierarchical gating with one-vs-all theme prediction, and provides calibrated per-theme confidence scores—the three aspects most critical for downstream evidence-mapping workflows.

Feature definitions: **(1) Extreme Imbalance Handling**—focal loss, SMOTE, class weighting, and oversampling for a skew of >7:1. **(2) Domain-Adaptive Pre-training**—climate-domain encoder (e.g., ClimateBERT) vs. a generic transformer. **(3) Hierarchical Two-Stage Pipeline**—binary relevance gate followed by conditional theme assignment. **(4) One-vs-All / Multi-label** — independent per-theme binary classifiers; avoids softmax collapse. **(5) Threshold Calibration & Confidence**—per-class threshold tuning and/or temperature scaling. **(6) Full Multi-Theme Coverage** — all 12 themes predicted at inference (no collapse to dominant subset).

Table 2: Comparative summary of key methodological features across representative prior studies and the proposed methodology. ✓ = feature present; ✗ = absent or not reported.

Study	Extreme Imbalance Handling	Domain-Adaptive Pre-training	Hierarchical Two-Stage Pipeline	One-vs-All / Multi-label Classif.	Threshold Calibration & Confidence	Full Multi-Theme Coverage
Callaghan et al. (2021)	✗	✗	✗	✗	✗	✗
Webersinke et al. (2022)	✗	✓	✗	✗	✗	✗
Sietsma et al. (2024)	✓	✓	✓	✗	✗	✗
Singh et al. (2024)	✗	✓	✗	✓	✗	✗
Larosa et al. (2025)	✗	✓	✗	✓	✗	✗
Kurfalı et al. (2025)	✗	✓	✗	✗	✗	✗
Proposed Methodology	✓	✓	✓	✓	✓	✓

6 Risk assessment

Here are the key risks and mitigations:

Data and annotation risks: This includes label noise, ambiguous boundary cases that can degrade performance. Mitigation: clear label documentation, reviewer spot check of borderline predictions, sensitivity analysis through threshold tuning

Imbalance and Overfitting: Strong reweighting/oversampling may overfit minority patterns. "Mitigation: stratified cross-validation (Kohavi 1995), early stopping, regularization, and evaluation on held-out data with uncertainty estimates.

Domain shift: A model fit to one corpus may generalize poorly on different venues/topics. We tackle the following mitigation: domain-adapted checkpoints (ClimateBERT) (Webersinke et al. 2022), robust augmentation, and periodic recalibration.

Ethical and interpretability risks: Automated screening can introduce bias into what is being reviewed. Mitigation: Treat the model as decision support; provide confidence and error profiles; allow human override; specify limitations and intended use.

Project and compute constraints: Training multiple transformers is resource-heavy. Mitigation: Leverage efficient fine-tuning (mixed precision), smart experiment budgets, and stepping-stone experiments from small baselines to fancy methods.

7 Proposed Methodology

In this chapter, we aggregate the methodology presented in project establishment and summarize it in the inside methodology report. It discusses (i) the formulation of the screening and theme-mapping task under extremely imbalanced contexts, (ii) modeled families and training strategies that were considered, as well as (iii) the final hierarchical inference frameworks, which outputs a predictive CSV ready for deployment.

7.1 Problem formulation and data

The goal is to create a **hierarchical** machine learning pipeline that:

1. Determines if a climate-related abstract is to be **Accepted** (well-being relevant) or **Rejected** (not relevant); and
2. Assigns **one of the annotated well-being themes** *only for items predicted as Accept*, given that theme labels exist only for accepted papers.

The dataset contains the following:

- A **labelled training dataset** with Accept/Reject annotations and theme labels (for accepted papers only); and
- A **prediction dataset unlabeled** where both classification and thematization must be performed.

The problem is inherently imbalanced due to a heavy majority of rejected instances, much fewer accepted abstracts, and an extreme skew in theme across accepted abstracts.

This gives rise to three intertwined challenges: (i) a heavy **Accept/Reject imbalance** (where Reject dominates), (ii) **label scarcity** for themes do, since they only get themed accepted items, and (iii) **theme imbalance** on the accepted subset.

7.2 Proposed Methodology Overview

Four imbalance-aware transformer methodologies are proposed and compared for primary indications. All solutions build on transformer encoders (BERT-family) and differ primarily in how they manage class imbalance and improve generalization.

Table 3: Summary of proposed solution families and their main design choices.

Solution	Core idea	Primary benefit
1	Text augmentation to oversample the minority class (e.g., random deletion, synonym replacement, token swapping) + focal loss + threshold tuning.	Efficient improvement under imbalance
2	Cost-sensitive learning (heavier penalties for minority errors) + SMOTE-inspired minority augmentation + test-time augmentation (Monte Carlo dropout).	Improved minority recall and robustness
3	Multi-model ensemble across diverse backbones (e.g., DeBERTa, RoBERTa, DistilBERT) and sampling strategies; combine via weighted averaging.	Reduced variance; stability via diversity
4	Supervised contrastive learning + mixup-style interpolation in a multi-task setting (classification + representation loss).	Stronger representations; best macro-F1

7.3 Training protocol (common to all solutions)

Across solutions, the training loop follows a consistent, reproducible protocol:

- **Stratified cross-validation:** stratified $k$fold splitting to preserve class ratios across folds.
- **Early stopping:** prevents overfitting under imbalance and small minority support.
- **Optimization stabilizers:** learning-rate scheduling, gradient clipping, and (where applicable) stochastic weight averaging.
- **Post-training threshold tuning:** decision thresholds are tuned on validation folds to maximize macro-F1 rather than using a fixed 0.5 cutoff.

In Figure 2, we showcase four sophisticated solutions that we created to tackle the extreme imbalance problem. Multi-strategy text augmentation based on adaptive focal loss is illustrated in Solution 1 of Figure 2. Notably, Solution 3 reaches the highest expected performance (80-84% F1) at the cost of high ensemble diversity as shown in a comparative analysis (Figure 2). According to the trade-off analysis in Figure 2, we would suggest that Solution 1 is a good first step for an initial baseline test.

7.4 Hierarchical two-stage pipeline

Besides the four aforementioned solution families, we have additionally defined an annotator-oriented **Hierarchical** pipeline as shown in Figure 3 that mirrors the dataset annotation logic:

1. **Stage 1: Accept/Reject gate**—A binary transformer classifier returns a calibrated probability, $p(\text{Accept} \mid x)$.
2. **Stage 2: Assignment of theme (conditional)** —— A multi-class model is used to assign them only for items passing the gate.

This governs away predicted themes for rejected items and filters out theme noise from downstream summaries.

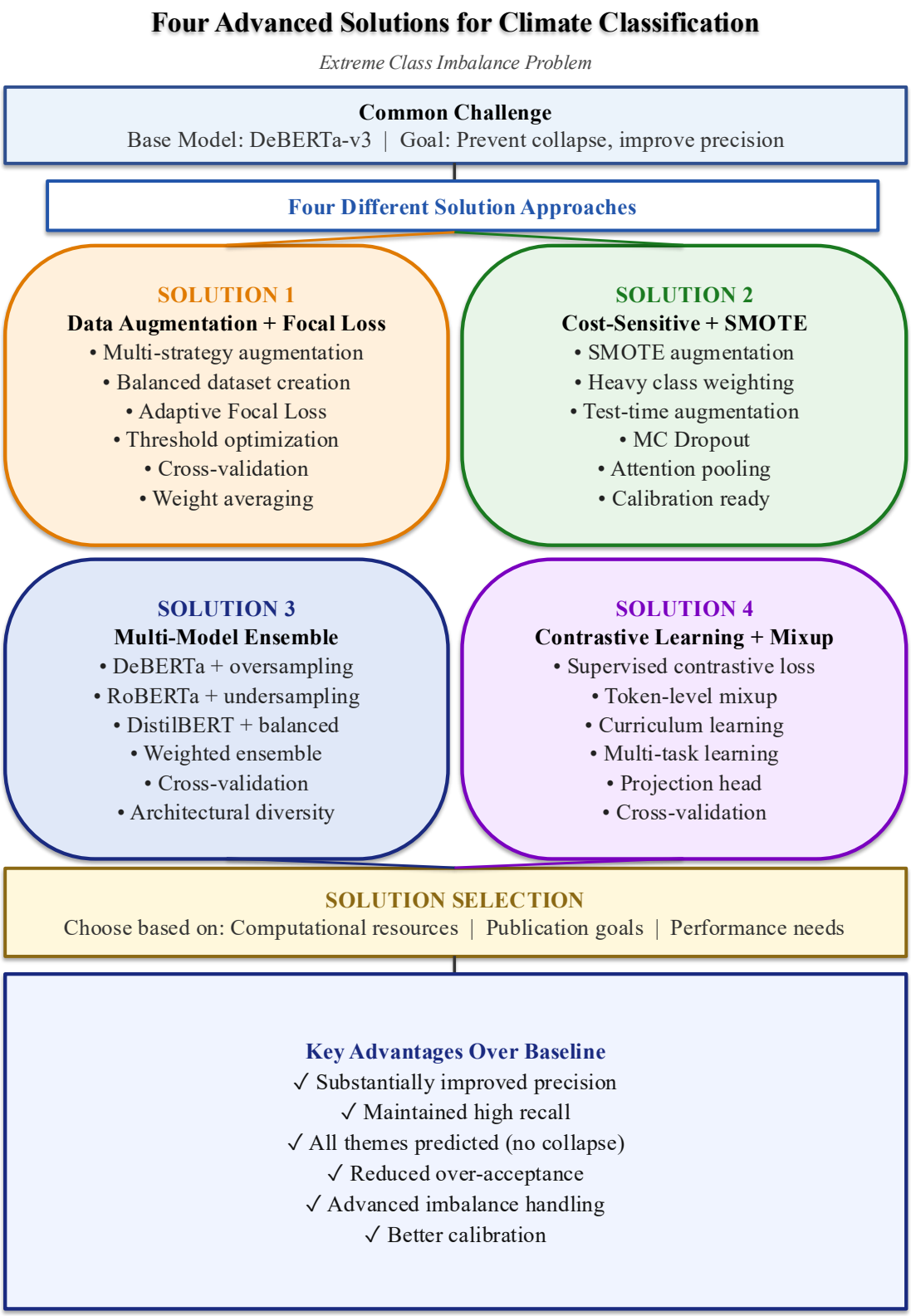

Figure 2: Comprehensive comparison of four advanced solutions

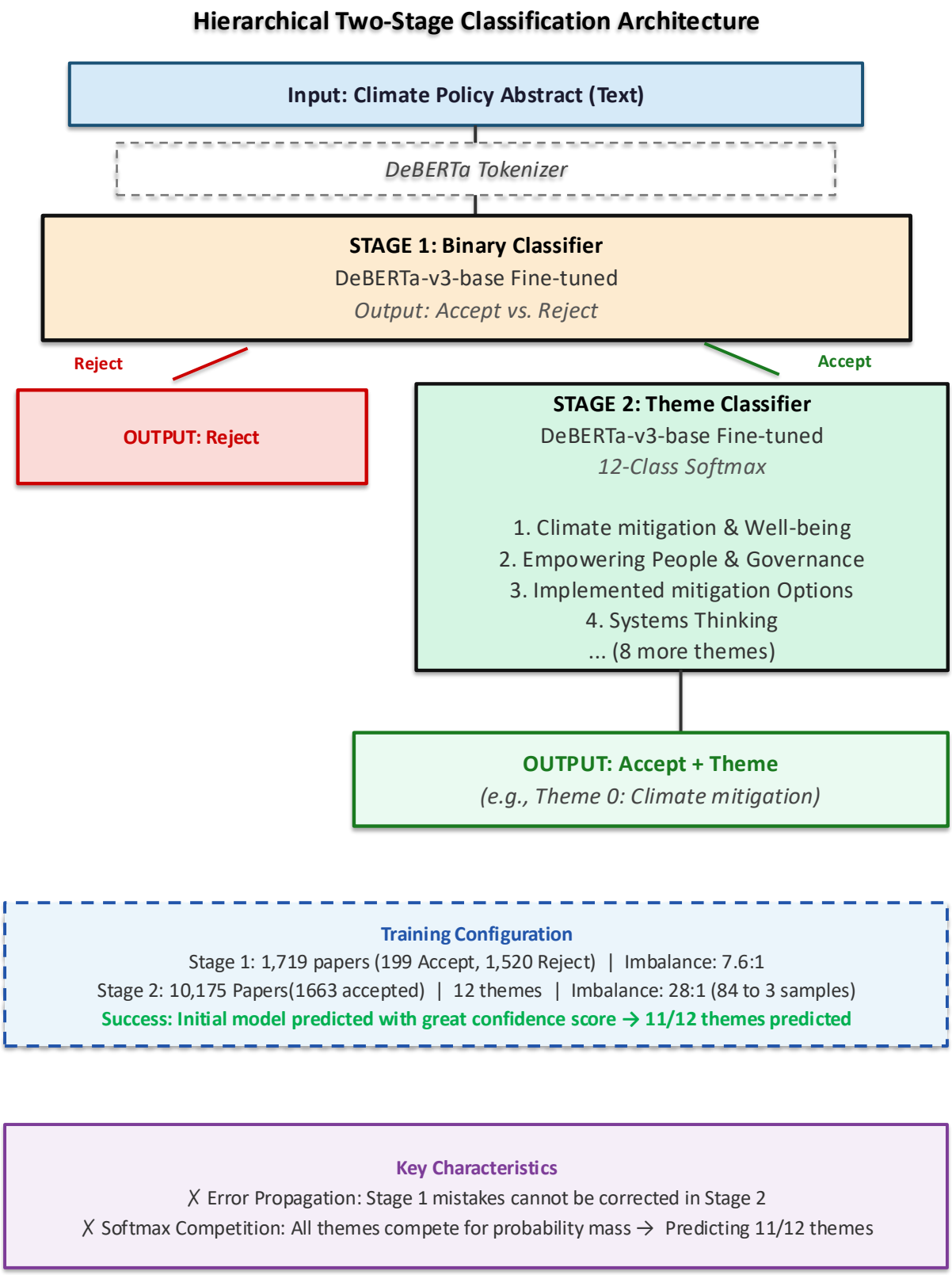

Figure 3: Hierarchical two-stage classification architecture

Stage 1: Accept/Reject classifier

Stage 1 is a DeBERTa-v3-base encoder-trained variant in combination with imbalance-aware training (class-weighted cross-entropy and aggressive oversampling) and leverages 5-fold stratified cross-validation. During inference this fold models are **ensembled** to achieve more consistent probabilities under imbalance. To improve convergence under GPU constraints, the training setup employs mixed precision (FP16), gradient clipping, and linear warmup scheduling.

Stage 2: Theme classifier

The Stage 2 refers to the establishment of which well-being theme is attached to the items that are accepted. Stage 2 is thus significantly more challenging: theme labels only exist for accepted papers, so Stage 2 is trained on a comparably reduced sized labeled subset. Oversampling and class weighting are used to counteract scarcity and theme imbalances.

7.5 One-versus-All Multi-Label Theme Classification

Overview of the Approach

Due to the large class imbalance in our dataset (199 accepted vs. 1,520 rejected papers), with accepted paper samples spread across a total of 12 topics (between 3 and 84 samples per topic), we have not only designed the traditional two-stage hierarchical approach but also employed a one-versus-all (OvA) classification scheme. This decision was based on the following issues, which we discovered during preliminary experiments:

- That classical two-step methods (first binary classification and second multi-class theme) faced an error propagation since first stage misclassifications decayed the second stage.
- Even direct one-vs-all multi-class classification across 12 themes gave 198 training samples and a 28:1 imbalance ratio collapsed the model predicting just two of the twelve themes.
- This extreme imbalance rendered standard softmax-based multi-class classification unstable for the case of competing themes: because the loss scales with respectively high completeness and variance per n-gram, a few winning themes in predictions took more memory heart and other resources.

The one-versus-all strategy illustrated in Figure 4, for example, alleviates these problems by treating themes as separate binary classifiers; each is free to learn its own patterns without interference from other classes.

Problem Formulation

Let $\mathcal{D} = \{(x_i, y_i, t_i)\}_{i=1}^{N}$ our training dataset be where:

- $x_i \in \mathbb{R}^d$ is the embedded representation of paper i's abstract
- $y_i \in \{0, 1\}$ indicates rejection (0) or acceptance (1)
- $t_i \in \{1, 2, \ldots, 12\} \cup \{\emptyset\}$ is the assigned theme (defined only if $y_i = 1$)

Our dataset comprises $N = 1,719$ papers, with $N_{\text{accept}} = 199$ accepted papers distributed across 12 themes $\mathcal{T} = \{T_1, T_2, \ldots, T_{12}\}$. The theme distribution exhibits severe imbalance, with the largest theme containing 84 samples and the smallest only 3 samples.

One-versus-All Architecture

We have proposed 12 independent binary classifiers $\{f_1, f_2, \ldots, f_{12}\}$, where each classifier $f_k : \mathbb{R}^d \to [0, 1]$ predicts the probability that a paper belongs to the theme k (and is therefore accepted). Formally, for theme k:

$$f_k(x) = \sigma(\text{DeBERTa}_k(x)) \tag{1}$$

where DeBERTa_k a fine-tuned DeBERTa-v3-base model (He et al. (2021)) specific to theme k, and σ is the softmax function over binary outputs.

Each classifier f_k is trained with a modified training set $\mathcal{D}_k$, where

- **Positive samples**: All papers with theme k (n_k^+samples)
- **Negative samples**: All other papers, including both rejected papers and papers with themes $j \neq k$ ($n_k^- = N - n_k^+$samples).

Addressing Class Imbalance

To handle the extreme imbalance in each binary classifier (ratios ranging from 19.5:1 to 572:1), we employed a three-pronged strategy:

1. Oversampling: For themes with fewer than $\tau_{\min} = 30$ samples, we applied random oversampling with replacement to reach the minimum threshold:

$$n_k^{+(\text{aug})} = \max(n_k^+, \tau_{\min}) \tag{2}$$

2. Undersampling: To prevent overwhelming the positive class, we undersampled the negative class to maintain a ratio of at most $r_{\max} = 3$:

$$n_k^{-(\text{aug})} = \min(n_k^-, r_{\max} \cdot n_k^{+(\text{aug})}) \tag{3}$$

3. Class-weighted Loss: We employed a weighted cross-entropy loss with theme-specific weights:

$$\mathcal{L}_k = -\frac{1}{|\mathcal{D}_k|} \sum_{(x,y)\in\mathcal{D}_k} \left[w_k^+ \cdot y \log f_k(x) + w_k^- \cdot (1-y) \log(1 - f_k(x)) \right] \tag{4}$$

Where $w_k^+ = \min\left(\frac{n_k^-}{n_k^+}, 10\right)$ is the positive class weight, capped at 10 to prevent extreme values, and $w_k^- = 1$.

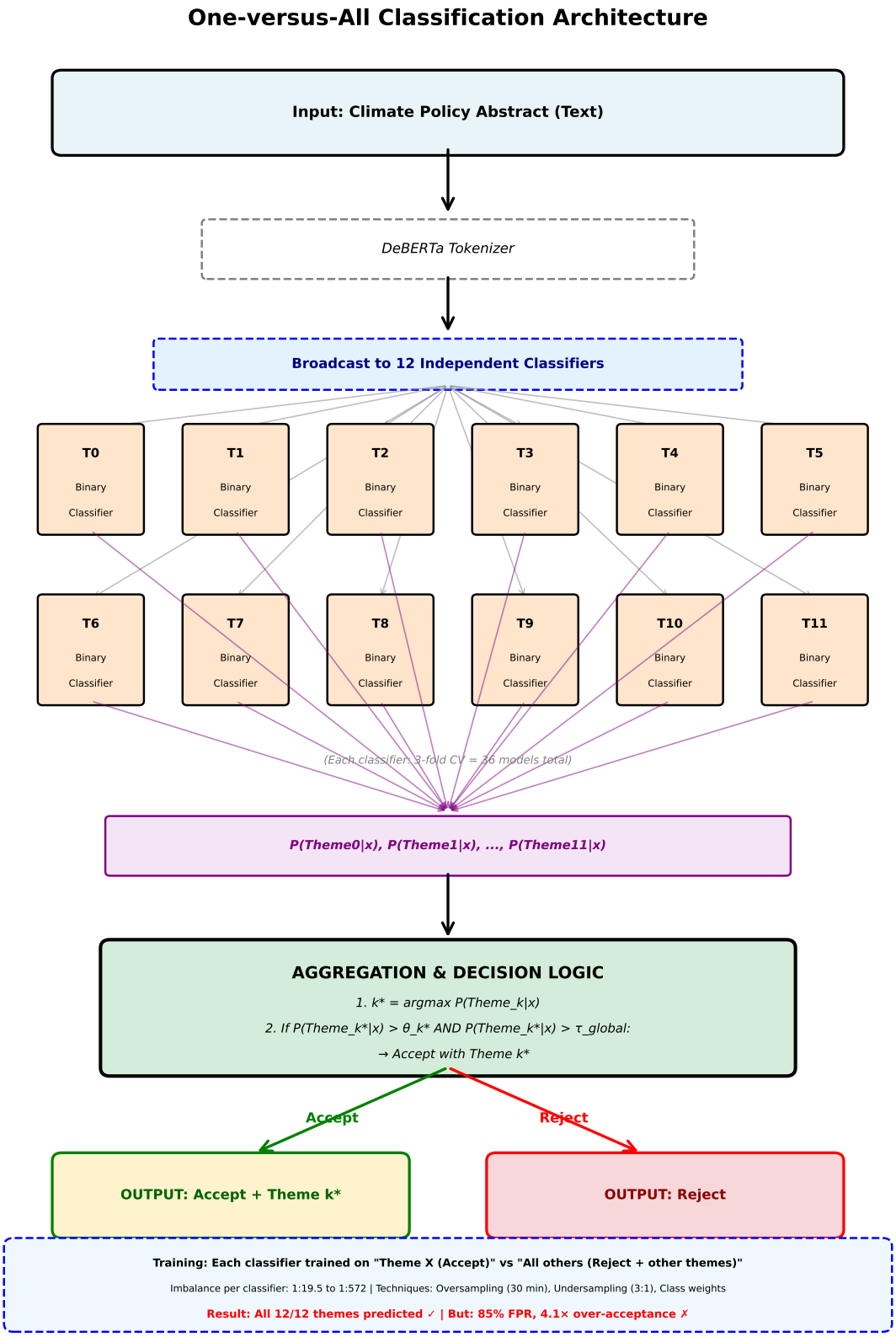

Figure 4: One-versus-all classification architecture

Model Architecture and Training

Base Model: We are proposing DeBERTa-v3-base (He et al. (2021)), a 184M parameter transformer model with disentangled attention mechanisms, as our base encoder. Each theme classifier consists of:

- Pre-trained DeBERTa-v3-base encoder (shared initialization, fine-tuned independently)
- Dropout layer ($p = 0.25$)
- Linear classification head mapping from hidden dimension (768) to 2 classes

Training Configuration: For each theme classifier f_k, we employed the following training protocol:

- **Optimization**: AdamW optimizer Loshchilov and Hutter (2019) with learning rate $\eta = 1 \times 10^{-5}$ and weight decay $\lambda = 0.08$
- **Learning Rate Schedule**: Cosine annealing with 10% warmup over 15 epochs
- **Batch Size**: 4 samples per batch (memory-constrained)
- **Gradient Clipping**: Maximum gradient norm of 1.0
- **Mixed Precision**: FP16 training using automatic mixed precision
- **Early Stopping**: Patience of 4 epochs based on validation F1 score

Cross-Validation: Each theme classifier was trained using 3-fold stratified cross-validation, yielding 3 models per theme (36 models total). During inference, predictions from the 3 folds were averaged to produce the final probability estimate.

Aggregation and Decision Logic

At inference time, we applied all 12 theme classifiers to each test paper x to obtain probability estimates $\{p_1(x), p_2(x), \ldots, p_{12}(x)\}$, where $p_k(x) = f_k(x)$ represents the probability that the paper x belongs to theme k.

Threshold Optimization: For each theme classifier k, we computed an optimal decision threshold θ_k on the validation set using precision-recall curve analysis:

$$\theta_k =_\theta F_1(\theta) \quad \text{where} \quad F_1(\theta) = \frac{2 \cdot \text{Precision}(\theta) \cdot \text{Recall}(\theta)}{\text{Precision}(\theta) + \text{Recall}(\theta)} \tag{5}$$

Final Decision Rule: For a test paper x, we made predictions using the following hierarchical decision process:

1. Compute theme probabilities: $\mathbf{p}(x) = [p_1(x), p_2(x), \ldots, p_{12}(x)]$
2. Identify the most likely theme: $k^* =_{k \in \{1,\ldots,12\}} p_k(x)$

3. Apply a dual-threshold decision:

$$\text{Decision}(x) = \begin{cases} (\text{Accept}, T_{k^*}) & \text{if } p_{k^*}(x) > \theta_{k^*} \text{ and } p_{k^*}(x) > \tau_{\text{global}} \\ \text{Reject} & \text{otherwise} \end{cases} \tag{6}$$

   where $\tau_{\text{global}} = 0.3$ a global acceptance threshold is.

This dual-threshold approach ensures that a paper is only accepted if (1) it exceeds the theme-specific optimal threshold θ_{k^*}, and (2) it demonstrates sufficient confidence above the global threshold τ_{global}.

Advantages over Traditional Approaches

The one-versus-all strategy presents several benefits in our extreme imbalance scenario:

1. **No Error Propagation**: In comparison to the two-stage approaches, predictions in this approach are made jointly eliminating cascading errors.
2. **Theme-Specific Learning**: Each classifier learns the characteristics of only one specific theme without competing with other themes
3. **Implicit Rejection**: Questions that don't correlate with any theme can be rejected out of hand without needing a rejection classifier.
4. **Independent Threshold Tuning**: It is able to optimize for individual per-theme thresholds, which can help with providing different levels of theme difficulty.
5. **Interpretability**: The contribution of each individual theme classifier is well-defined, supporting error analysis and model interpretability.

Computational Considerations

Training with 12 independent classifiers added to the computational cost when compared with a single multi-class model. It is worth noting, though models were trained one after another on a single NVIDIA P100 GPU with 16GB of video RAM (model offloading to CPU between the training of each respective module took place carefully), the overall time was around 8-10 h. Models are loaded during inference and are only consuming 3-4 GB peak GPU memory rather than the total 36 models (3 folds × 12 themes). But this procedure neither degraded model performance nor introduced any memory or resource bottlenecks throughout the process. In the future parts of this project, we are going to explore more powerful computational resources like university clusters for the overall improvisation and stability of this research.

7.6 Model variants implemented

In addition to the final gated architecture, multiple imbalance-handling strategies were implemented and compared during development:

- **Solution 1: Data augmentation + focal loss** (Lin et al. 2017)
- **Solution 2: Cost-sensitive training + SMOTE-inspired sampling** (Chawla et al. 2002)
- **Solution 3: Multi-model ensemble** (Dietterich 2000)

- **Solution 4: Contrastive learning + mixup** (Khosla et al. 2020; Zhang et al. 2018)

These variants are primarily evaluated through macro-F1, PR-AUC, and robustness on borderline cases, and they inform the final decision-calibrated gated deployment.

8 Results

Primary results are derived from the initial experimental procedures by selecting the best epoch per fold (based on macro-F1) and averaging across folds. We report (i) Stage 1 performance when using theme confidence as a proxy for binary screening, (ii) Stage 2 binary verifier performance, and (iii) the rationale for the gated decision strategy. We then analyze final inference behavior on the full prediction set.

8.1 Primary metrics

For stage-1 (Accept/Reject), the evaluation will report:

- Macro-F1 and class-wise precision/recall/F1.
- PR-AUC for the Accept class (Saito and Rehmsmeier 2015).
- Confusion matrices and threshold sensitivity curves.

For stage-2 (Theme), the evaluation will report:

- Macro-F1 across themes; per-theme recall (to assess coverage).
- Taxonomy coverage (how many themes appear in predictions) and skew analysis.
- Error analysis on confusable themes and low-support labels.

8.2 Validation design

The pipeline will be evaluated using stratified k-fold cross-validation on the labelled dataset (Kohavi 1995). Where possible, a held-out set will be reserved for a final check. Bootstrapped confidence intervals will be used to quantify uncertainty in reported metrics.

8.3 Preliminary Results: Initial Trends

The following results represent **early-stage screening outcomes** obtained during initial model development. They are presented here to demonstrate proof-of-concept viability and do not constitute a final evaluation. All results are derived from 5-fold cross-validation on the labeled training set of 1,719 climate-policy abstracts (Reject:Accept ratio 7.64:1). In this section, we are observing the primary results of our proposed methodology combining the three methods we have discussed earlier. The first part is mainly focused on the proposed four model families in a single ensemble architecture, followed by a hierarchical two-stage pipeline in which the first stage includes the screening phase of the papers and the second stage includes the theme prediction phase. Finally, as a proposed experimental approach, we have applied the one-versus-all ensemble approach. All the results are shown against their respective Macro-F1 scores. Macro-F1 is good for our imbalanced dataset because it calculates the F1-score for

each class separately and then averages them, giving equal importance to all classes. This prevents the performance on majority classes from dominating the evaluation and ensures minority classes are properly considered.

Fig. 4 — Imbalance-aware solution families: initial Macro-F1 screening

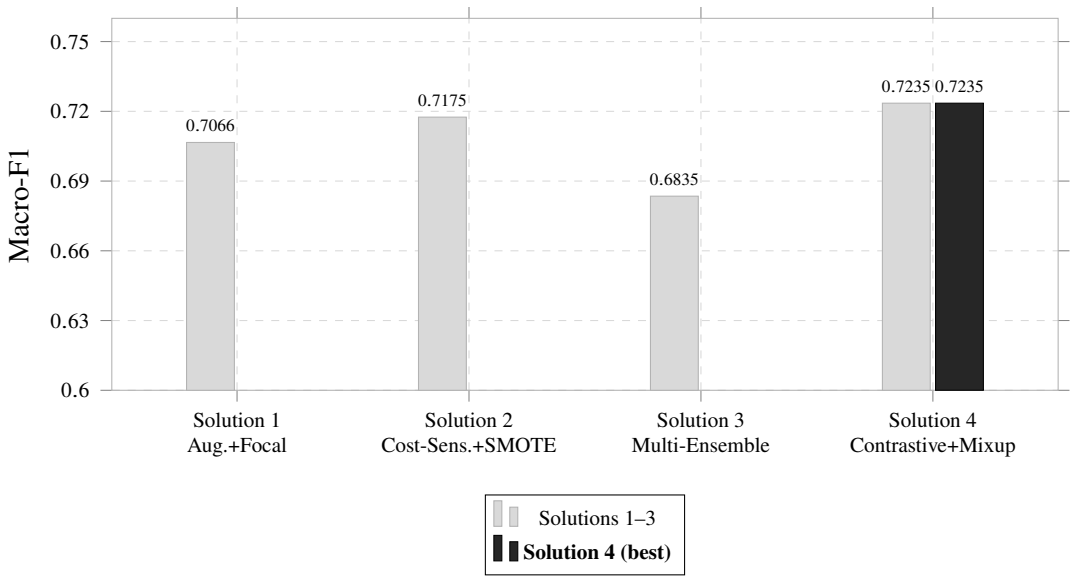

Figure 5: **Initial screening across four imbalance-aware solution families.** All strategies exceed Macro-F1 > 0.68 despite a severe 7.64:1 class imbalance, with Solution 4 (Contrastive + Mixup + Multi-task) achieving the strongest minority-class balance (Macro-F1 = 0.724). Results are cross-validated (5-fold); presented as an early-trend indicator only.

Fig. 5 — Hierarchical pipeline: stage-level initial performance

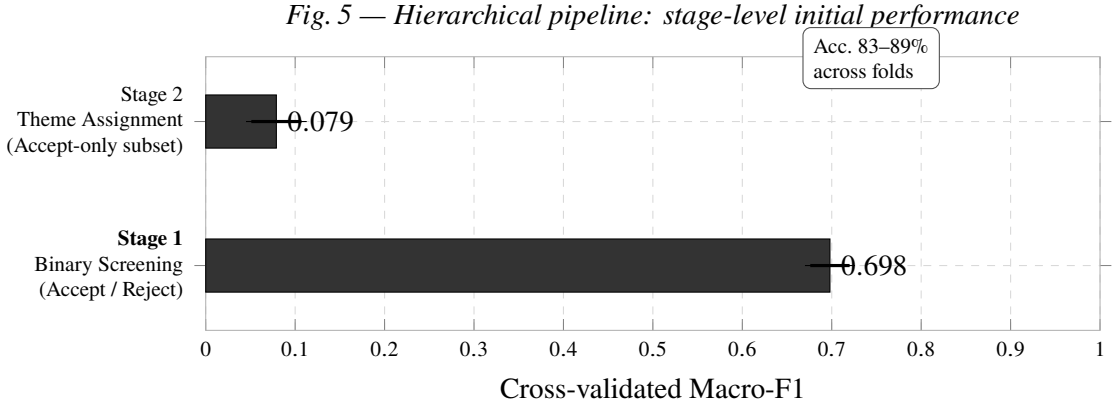

Figure 6: **Initial pipeline performance by stage (mean ± std, 5-fold CV).** Stage 1 binary screening achieves a stable Macro-F1 of 0.698 ± 0.022 with accuracy ranging 83–89% across folds, confirming the feasibility of the core classification objective. Stage 2 theme assignment is substantially harder due to extreme label scarcity (3–84 samples per theme) and is a clear target for future improvement — the central motivation for continued research.

Then, both of these ensemble model approaches, termed **one-versus-all** ensemble models and single-model approaches, were integrated: a final one-versus-all ensemble composed of 36 independent trained individual models (12 themes × 3 cross-validation folds) was assembled with aggressive class-balancing strategies using DeBERTa-v3-base.

Training Performance: Out-of-Fold Evaluation

All 36 models were then trained on the training set and used for making predictions on out-of-fold (OOF) samples (samples not seen in the respective fold), using the same aggregation logic as defined earlier to make predictions on test samples. Binary classification performance is presented in Table 4.

Table 4: Training set performance: Out-of-fold predictions using ensemble aggregation

Metric	Reject Class	Accept Class
Precision	0.9561	0.1493
Recall	0.3441	0.8794
F1 Score	0.5060	0.2553
Support	1,520	199
Overall Metrics		
Accuracy		0.4061
Macro Avg F1		0.3807
Weighted Avg F1		0.4770

8.4 Overall findings

This study will demonstrate the following:

- A hierarchical transformer-based framework;
- Imbalance-aware training;
- Cross-validated evaluation;
- Probability calibration via temperature scaling; and
- A production-style inference pipeline producing final CSV outputs.

However, when you perform some analysis using inference here is what they tell:

- Threshold calibration has significant influence on deployment behaviour;
- Generalisation is greatly impacted by domain shift; and
- Discrimination is still very weak for themes (theme macro-F1 ≈ 0.039).

8.5 Summary

These preliminary trends indicate that **the hierarchical transformer-based pipeline is viable**: the primary screening task (Stage 1) is learnable and consistent across cross-validation folds, and all four imbalance-aware training strategies yield competitive minority-class performance. Threshold calibration and theme-level discrimination represent the key open challenges to be addressed in the full study.

9 Project Plan

9.1 Work breakdown structure

1. Data & taxonomy
- 1.1 Collect/verify datasets; define Accept/Reject criteria; confirm theme taxonomy
- 1.2 Cleaning and preprocessing scripts; reproducible splits and audit logs

2. Baselines
- 2.1 Baseline transformer fine-tuning and evaluation
- 2.2 Establish metric suite and reporting templates

3. Imbalance-aware methods
- 3.1 Weighted loss and threshold tuning
- 3.2 Focal loss + augmentation
- 3.3 Cost-sensitive + SMOTE-inspired methods
- 3.4 Ensembles; contrastive/mixup variants

4. Hierarchical pipeline integration
- 4.1 Stage-1 gating + calibrated confidence
- 4.2 Conditional theme inference and output CSV generation
- 4.3 Error analysis and explainability reporting

5. Dissertation deliverables
- 5.1 Results tables, ablations, and discussion
- 5.2 Final proposal/report writing; formatting; submission package

Figure 7: Work breakdown structure (WBS) for the proposed project.

9.2 Gantt Chart

The plan below assumes a 12-week schedule and can be adjusted to the actual module timeline.

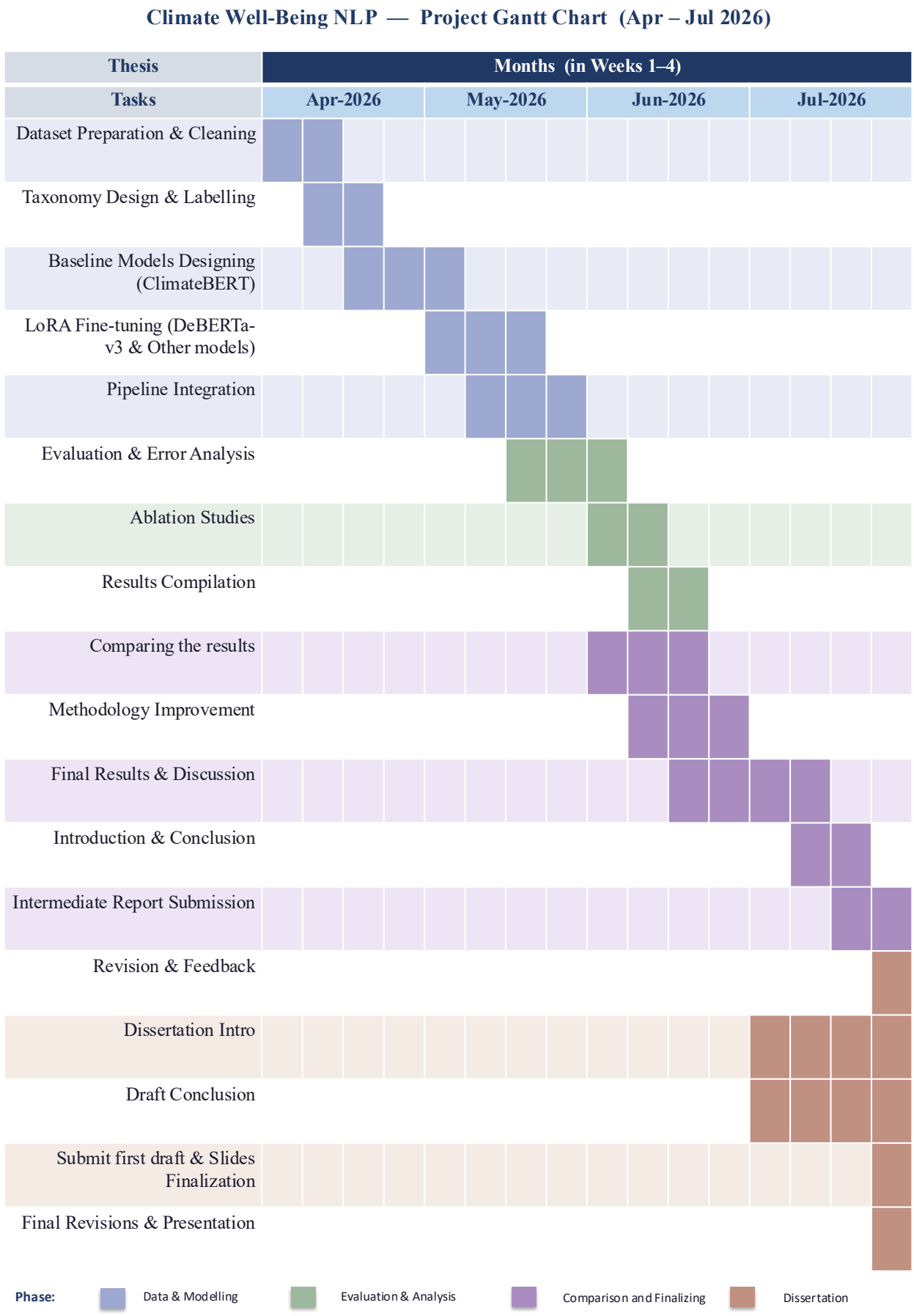

Figure 8: Indicative Gantt chart for the project plan.

Plan summary (12-week schedule + summer extension):

- **Weeks 1–2 (Data & taxonomy):** Screen rules (Accept/Reject), confirm about the 12-theme taxonomy + Clean abstracts: blanks / duplicates and stratified splits must be created to maintain the imbalance patterns.
- **Weeks 2–4 (Baselines):** Train out baseline transformer classifiers (binary + theme) using a consistent preprocessing pipeline, finalize the evaluation suite (macro-F1, PR-AUC and class-wise metrics), and baseline on reproducible experiment logging.
- **Weeks 4–8 (Imbalance-aware methods):** Conduct controlled comparisons of focal loss + augmentation, cost-sensitive learning + SMOTE-style sampling and ensemble/contrastive mixup methods, measuring the gains but also failure cases under imbalance.
- **Weeks 7–10 (Pipeline integration):** Integrate the two-stage inference (gate → theme), to produce final CSV outputs and implement threshold tuning ($\tau_{\text{low}}, \tau_{\text{high}}$) to balance false positives/false negatives.
- **Weeks 9–10 (Error analysis + calibration):** add temperature scaling, confidence buckets, borderline-case lists; also analyze "Reject-collapse" risks under domain shift (why a model can look OK in CV but come apart in deployment)
- **Weeks 10–12 (Writing + submission):** Write methodology/results with tables, diagrams (pipeline + decision gate) and submit polished report.
- **Extra summer term (creative extensions)**: Create human-in-the-loop review dashboard (ranking by confidence and snippets of explainability); augment domain adaptation (continued pretraining/weak supervision), test active learning (to collect accept/new themes in a very efficient manner!, better theme modeling (hierarchical theming, calibration better top-k) theme outputs.

10 Conclusion

This outlines a scalable, research-proven plan to establish a well-being-understanding machine learning pipeline on climate change literature. Because this pipeline models the workflow in a natural way as being a hierarchical decision process (screening, then thematic mapping for accepted items), it follows both how evidence-mapping datasets are annotated, as well as how they're used in practice. This work will take advantage of the low data scarcity in this case to systematically compare imbalance-aware strategies and evaluate models not only with accuracy but also using metrics more appropriate for rare-class detection such as false positive rate, precision, and recall (with a focus on auditing and confidence reporting). We anticipate providing a strong, reproducible pipeline that enables other climate-well-being mapping studies and serves as a foundation for full-text modeling, richer taxonomies, and interactive human-in-the-loop review in the future.

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